

Sumit Srivastava

Curriculum Vitae

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Professional Summary

- 4 years of academic research experience in Artificial Intelligence, focusing on Natural Language Processing (NLP) and Human Computer Interaction (HCI).
- Experience in defining research goals and executing them.
- 8 years of experience in building large-scale enterprise software applications.

Work Experience

February 2025 **Senior Backend Engineer**, *Tibo Energy*, The Netherlands.

- Present ○ Responsible for designing, implementing, and maintaining features for an industrial energy management software.

Main skills and technologies: Typescript, NodeJS, MySQL, Cassandra

2020–2024 **Doctoral Researcher (defense pending)**, *University of Twente*, The Netherlands.

Topic: Interactive personalization of explanations in conversational agents.

- Conducted NLP research on personalization in conversational agents using lexical alignment, grounded in psycholinguistics and cognitive science.
- Designed and implemented conversational agents for generating interactive, personalized textual explanations.
- Conducted empirical user studies evaluating trust, persuasiveness, cognitive load, and information uptake in human–AI dialogue.
- Researched dialogue generation methods using Large Language Models for controllable lexical alignment (paper to be submitted to ACM UMAP 2026).
- Supervised eight MSc and BSc students.
- Published peer-reviewed work at ACM IUI, ACM UMAP (short), and Computer Speech & Language, and co-authored additional papers and posters through student supervision.

Main skills and technologies: NLP, Natural Language Generation, Large Language Models (LLMs), GenAI, Conversational AI, Explainable AI, Personalization, Dialogue Systems, User Studies, Experiment Design, Scientific Writing, Cognitive Science, Psycholinguistics, Human–Computer Interaction.

Advisors:

Dr. Mariët Theune, *Associate Professor, Human-Media Interaction Lab, EEMCS*, University of Twente

Dr. Alejandro Catalá Bolós, *Associate Professor, CiTIUS*, University of Santiago de Compostela

- Sep, 2019 – **Master of Science researcher in Artificial Intelligence**, *Katholieke Universiteit Leuven*, Belgium.
 May, 2020
 Topic: Metaphor interpretation (NLU) using Theory of Mind at scale.
 Main activities:
- Conducted research on automated interpretation of simple metaphors.
 - Used Bayesian inference (Rational Speech Act model) and a large-scale concept map **ConceptNet**.
 - Proposed and developed the research topic independently.
- Advisor:
Dr. Walter Schaeken, *Full Professor, Brain and Cognition, Faculty of Psychology and Educational Sciences, KU Leuven* ([Personal Web-page](#))
- 2013 – 2019 **Team Manager & Technical Lead**, *MiQ Digital*, India.
- Led and mentored a team of seven engineers while owning seven production services.
 - Revamped a real-time ad bidding platform handling 1.5M RPS, achieving sub-2 ms p99 latency via event-driven architecture and database optimizations.
 - Defined company-wide code and documentation standards and contributed performance improvements to an open-source PostgreSQL library.
- 2015 – 2016 **Senior Software Engineer**, *MiQ Digital*, India.
- Developed a real-time bid modification service processing up to 1.5M requests per second.
 - Modernized 70% of a core interface service through refactoring, library upgrades, and improved modularity.
 - Built a horizontally scalable system to compress and ship up to 90 TB of logs daily to AWS S3.
- 2013 – 2014 **Software Engineer**, *MiQ Digital*, India.
- Designed a data ingestion service supporting 12+ protocols and 30+ heterogeneous data sources.
 - Implemented custom processing pipelines storing data in AWS S3 and Redshift.
 - Mentored new engineers and contributors during and beyond this role.
- July 2012 – **Software Engineer / Associate Software Engineer**, *Apostek Softwares Pvt. Ltd.*,
 August 2013 India.
- A mobile advertising platform and a central point currency system where users earn points to watch ads and redeem them for various rewards.
 - Main activities: Critical enhancements, bug fixes, and addition of new features across different portals concerned with the project using Ruby on Rails as the core development framework integrated with Cassandra, MySQL for handling database activities and HTML, CSS, JavaScript for UI enhancements.

Education

- 2020–2024 **Doctor of Philosophy (defense pending), Computer Science**, *University of Twente*, The Netherlands.

- 2019–2020 **Master of Science, Artificial Intelligence**, *Katholieke Universiteit Leuven*, Belgium.
Grade: Cum Laude
Courses taken: Machine learning, Uncertainty in Artificial Intelligence, Cognitive Science, Computer Vision, Artificial Neural Networks and Deep Learning, Knowledge Representation (First Order Logic), Support Vector Machines, Fundamentals of Artificial Intelligence, Neural Computing
- 2010–2012 **Master of Technology, Information Technology**, *Indian Institute of Information Technology, Allahabad*, India.
Courses taken: Computational Intelligence, Information Retrieval, Computer Vision, etc.
CGPA: 9.2/10
- 2006–2010 **Bachelor of Technology, Computer Science**, *Uttar Pradesh Technical University, Lucknow*, India.
 Operating Systems, Database Management Systems, Networks, Digital Signal Processing, Digital Image Processing, Finite State Automata
Percentage: 71/100

Fellowships & Achievements

- 2020 – **Marie Skłodowska-Curie fellowship** as part of the EU-funded project Interaction present Natural Language Technology for Explainable Artificial Intelligence (NL4XAI).
- 2010 **Graduate Aptitude Test Examination** (GATE, India) rank 3623

Skills

- Research skills: Scientific Writing, Quantitative Research, Data Interpretation, Research design, Qualitative Research, Statistical analysis, Research Planning, Research presentation
- Programming Languages: Java, Python
- Artificial Intelligence: Artificial Neural Networks, etc.
- Artificial Intelligence Tools: Rasa (chatbot), Google DialogFlow, PyTorch, Keras, spaCy, NTLK
- Web servers: Nginx, Apache Tomcat, Vert.x
- Cloud: AWS, Appnexus(Xandr), Google Cloud
- Databases: MySQL, PostgreSQL, MongoDB, AWS Redshift
- CVS: Git
- Operating System: Linux

References

- **Dr. Mariët Theune** (co-supervisor)
Associate Professor, HMI, EEMCS, University of Twente, The Netherlands.
Email: m.theune@utwente.nl
- **Dr. Alejandro Catala** (co-supervisor)
Assistant Professor, CiTius, University of Santiago de Compostela, Spain.
Email: alejandro.catala@usc.es
- **Prof. Dr. Chris Reed** (secondment supervisor)
Professor, Center for Argumentation Technology, University of Dundee, Scotland, The United Kingdom
Email: chris@arg.tech

Research Publications

Journal Articles

- 2025 **Sumit Srivastava**, Mariët Theune, Alejandro Catala, and Chris Reed., 2025, Your Word Against Mine: Lexical Alignment in Argumentative Dialogues with a Conversational Agent (**Under Review**), *Cognitive Science*.
- 2024 **Sumit Srivastava**, Suzanna Wentzel, Mariët Theune, and Alejandro Catala., 2024, Measuring and Implementing Lexical Alignment: A Systematic Literature Review, *Computer Speech & Language*, 101731, Elsevier.

In Conference Proceedings

- 2026 **Sumit Srivastava**, Mariët Theune, and Alejandro Catala. Weighted decoding for controlled lexical alignment in llm-based dialogues (**Under Review**). In *Proceedings of the 34rd ACM Conference on User Modeling, Adaptation and Personalization (UMAP '26)*. ACM, 2026.
- 2024 **Sumit Srivastava**, Mariët Theune, Alejandro Catala, and Chris Reed. Trust in a human-computer collaborative task with or without lexical alignment. In *Adjunct Proceedings of the 32nd ACM Conference on User Modeling, Adaptation and Personalization (UMAP Adjunct '24)*, July 1–4, 2024, Cagliari, Italy. ACM, New York, NY, USA, 9 pages. ACM, 2024.
- 2023 **Sumit Srivastava**, Mariët Theune, and Alejandro Catala. The role of lexical alignment in human understanding of explanations by conversational agents. In *Proceedings of the 28th International Conference on Intelligent User Interfaces*, pages 423–435, 2023.
- 2013 **Sumit Srivastava** and Ramesh Chandra Tripathi. Real time mono-vision based customizable virtual keyboard using finger tip speed analysis. In *Human-Computer Interaction. Interaction Modalities and Techniques: 15th International Conference, HCI International 2013, Las Vegas, NV, USA, July 21-26, 2013, Proceedings, Part IV 15*, pages 497–505. Springer, 2013.